Arun Bagavathiannan

(979) 575-3148 | Email | LinkedIn | GitHub | Portfolio

Education

The University of Texas at Austin Austin, TX
Bachelor of Science, Computer Science *Expected Spring 2028*

- Relevant Coursework: CS 311 (Discrete Math), CS 314 (Data Structures), ECE 312H (Software Design & Implementation I), ECE 319H (Embedded Systems), M 340L (Matrices and Matrix Operations)

Experience

Technical Lead, UT Registration Plus Jan. 2026 – Present
Longhorn Developers

- Promoted from Software Engineer to Technical Lead in Aug. 2026; leading a team of 5 developers on a Chrome extension serving 60,000+ students
- Ran end-to-end hiring for the developer fellowship: screened 80+ applications, interviewed finalists, and selected 4 engineers (~5% acceptance)
- Engineered a Course Status Checker that enables real-time course availability tracking within the calendar UI — cited in UT CS News coverage of the extension
- Designed a throttled refresh system in React and TypeScript to prevent API rate-limiting while maintaining compliance with university infrastructure constraints
- Built a reactive store architecture to synchronize state between background service workers and the frontend

Content Creator Jul. 2021 – Present
YouTube

- Grew YouTube channel to over 20,000 subscribers with 32M+ impressions and 7M+ views by shipping consistent, high-quality video editing content through After Effects and Alight Motion
- Launched and operated buyorphic.com, an e-commerce site selling editing presets, generating $500+ in sales over one year

ML Research Intern Jan. 2025 – Mar. 2025
Texas A&M University

- Presented results at 3 conferences including the Weed Science Society of America (Vancouver) and European Weed Research Society (Spain)
- Designed and trained a YOLOv8 object detection pipeline, taking a raw dataset of 100+ images to a production-ready model with 80%+ precision and recall
- Engineered automated data augmentation workflows using Python and OpenCV, scaling training data 6x (100 to 600+ images) and cutting manual processing time

Projects

Breakout Sep. 2026 – Present

- Built MV3 Chrome extension scoring YouTube videos against rolling channel medians, parsing ytInitialData instead of the quota-limited API to stay free
- Designed TF-IDF niche filter building weighted vocabulary from seed channels, surfacing adjacent outliers keyword search misses; shipped free against $13/mo incumbents

Skills

Programming: Python, Assembly, C, C++, TypeScript, Java
Tools/Platforms: ChromaDB, Chrome Extension (Manifest V3), React, Node.js, Git/GitHub
AI/ML: Gemma 4 E4B, LLM Inference, RAG Pipelines, Model Training/Validation, MCP, Claude API, Ollama
Languages: English (Native), Tamil (Native), German (Novice), Japanese (Novice)

Awards

1st Place, AI Data Portability Hackathon – The AI Collective, AITX, DTI Mar. 2026
International Education Fee Scholarship (IEFS) Recipient Mar. 2026
Friends of Alec Engineering Scholarship Recipient Feb. 2026